

REFLECTION FOR HW13B

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Design Rationale

For this exercise, I focused on applying the MVC (Model-View-Controller) architectural pattern correctly. I separated the project into three packages: model, view, and controller. This separation helps keep the source code modular, readable, and maintainable.

The DateModel class is responsible for handling all date-related operations using Java's LocalDate class. The DateView class handles all graphical user interface components, including the date display field, input field, buttons, and dialog messages. The DateController class acts as the connection between the model and the view by listening to user actions and updating the displayed date.

One important design decision was using LocalDate instead of manually calculating dates. This allows the application to correctly handle leap years, February 29, different month lengths, and other date-related edge cases automatically.

Main Functionalities

1. Display Current Date:

When the application starts, the current date is automatically displayed using a Vietnamese-style date format.

2. Date Manipulation:

Users can enter a number x and move the displayed date:

- x days before

- x days after
- x months before
- x months after

3. Input Validation:

The application validates user input and handles invalid cases such as empty input, non-numeric input, negative values, and excessively large numbers.

4. Error Messages:

If invalid input occurs, the application displays clear error messages using JOptionPane dialog boxes.

Implementation

The project follows the MVC structure clearly.

In the model layer, DateModel stores the current LocalDate object and provides methods for modifying the date by days or months.

In the view layer, DateView extends JFrame and creates all Swing GUI components such as JTextField, JButton, JLabel, and JPanel. Layout managers are used to organize components into a clean and user-friendly interface.

In the controller layer, DateController registers button listeners, validates user input, calls methods from the model, and updates the view accordingly.

Overall, this exercise helped me better understand the MVC pattern, event handling, Java Swing GUI programming, and safe date manipulation using Java's LocalDate API.